# Topological Properties, Optimal Routing, and Embedding on the $K$-valent Graph

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**Abstract**

*This paper introduces a new family of Cayley graphs, named the $k$-valent graph, for building interconnection networks. We show that the $k$-valent graph possesses many valuable topological properties, such as regular with the node-degree $k$, logarithmic diameter subject to the number of nodes for some fixed $k \geq 6$, and maximally fault tolerance. We also show that the $k$-valent graph contains trivalent graphs defined in* [*P. Vadapalli and P. K. Srimani, "Trivalent Cayley graphs for interconnection networks," Information Processing Letters, vol. 54, pp. 329–335, 1995.*] *as its subclass of graphs. An algorithm is proposed to determine a shortest path between arbitrary two nodes. Besides, embedding of cycles or cliques on the $k$-valent graph is also discussed.*

**Keywords:** *Cayley graphs, diameter, embedding, interconnection networks, shortest path routing.*

## 1 Introduction

Design of interconnection networks is an important issue of the parallel processing or distributed system. Many interconnection networks have been proposed in the literature [1, 2, 4, 5, 7, 6, 8, 9, 10, 11, 13, 15, 16, 19, 21]. The choice of network topologies significantly determines the performance of these kinds of systems. Several important measurements, such as connectivity, node-degree, diameter, symmetry, fault tolerance and so on, are discussed for building interconnection networks [24]. The interconnection network is often modelled as an undirected graph, in which the nodes correspond to the processors and the edges correspond to the communication channels. Thus the terms, graph and network, node and processor, edge and channel, are used interchangeably throughout this paper.

A class of Cayley graphs was shown to be very suitable for designing interconnection networks [2]. These graphs are based on permutation groups. There are quite a few well-known interconnection networks, including hypercubes [4], star graphs [1, 2], pancake graphs [2], and so on, which belong to Cayley graphs. All of these graphs are regular and symmetric. However, the degree of the nodes increases with the size of the graph. It can make the use of these graphs prohibitive for networks with large number of nodes. Therefore, the fixed node-degree networks are important from VLSI implementation point of view [16]; there are applications where the computing nodes in the interconnection network can have only a fixed number of I/O ports [5]. Therefore, designing of an interconnection networks to keep fixed node-degree when the number of nodes increases may be of interests by itself.

P. Vadapalli and P. K. Srimani [19] proposed the trivalent Cayley graphs with the fixed node-degree 3. This class of graphs was shown to be regular, to have logarithmic diameter, and to be maximally fault-tolerant. Several topological properties and shortest path routing algorithm were discussed in [14, 18, 19, 20, 22]. However, both the node-degree and the node-connectivity of trivalent graphs are limited by 3, and thus the probability of failure get increasing when the size of the network is extended. In this paper, we generalize the work of [19] to propose a new family of Cayley graphs, named the $k$-valent graphs $G_{k,n}$. It possesses a nice property that the degree of each node is fixed by $k$ without regarding the size of the network. We show that the trivalent graph is a subclass of the $k$-valent graph when $k = 3$. We also show that the $k$-valent graph have logarithmic diameter when $k \geq 6$, maximally fault tolerance, and can

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embed disjoint cycles or cliques. Besides, a shortest path routing algorithm which is more complicated than that of the trivalent Cayley graph is also presented.

The rest of this paper is organized as follows: In the next section, the $k$-valent Cayley graph is introduced. In Section 3, a shortest path routing algorithm is proposed. Based on the algorithm, the diameter is established in this section. Embedding and node-connectivity are discussed in Section 4 and Section 5, respectively. Finally, some concluding remarks are given in Section 6.

## 2 The $k$-valent Graph

The following is a formal definition of the $k$-valent Cayley graph.

**Definition 1.** A *$k$-valent graph $G_{k,n}$* is an undirected graph with $N = n(k-1)^n$ vertices for any integers $n \geq 2$ and $k \geq 3$. Each node $v$ of $G_{k,n}$ has the form $s_0 s_1 \ldots s_{m-1} \tilde{s_m} s_{m+1} \ldots s_{n-1}$ corresponding to a string of $n$ symbols selected from $\{0, 1, \ldots, k-2\}$ such that exactly one symbol $\tilde{s_m}$ is in *marked* form and the others are in *unmarked* form. We sometimes use $v^m$ for representing a node $v$ with the marked symbol on position $m$, and thus the notations $v^m$ and $v$ are used interchangeably throughout this paper. Let $s_i^* = s_i$ or $\tilde{s_i}$. Each edge is of the type $(v, \delta(v))$, where $\delta \in \{f, f^{-1}, g_1, g_2, \ldots, g_{k-2}\}$ is a *generator* defined as follows:

- $f(u^m) = v^{(m-1) \bmod n}$, where $u^m = s_0^* s_1^* \ldots s_{n-1}^*$, $v^{(m-1) \bmod n} = s_1^* s_2^* \ldots s_{n-1}^* \alpha^*$ and $\alpha = (s_0 + 1) \bmod (k-1)$;
- $f^{-1}(u^m) = v^{(m+1) \bmod n}$, where $u^m = s_0^* s_1^* \ldots s_{n-1}^*$, $v^{(m+1) \bmod n} = \beta^* s_0^* s_1^* \ldots s_{n-2}^*$ and $\beta = (s_{n-1} - 1) \bmod (k-1)$;
- $g_i(u^m) = v^m$, where $u^m = s_0^* s_1^* \ldots s_{n-1}^*$, $v^m = s_0^* s_1^* \ldots s_{n-2}^* \gamma^*$ and $\gamma = (s_{n-1} + i) \bmod (k-1)$ for $1 \leq i \leq k-2$.

For example, $\{\tilde{0}0, \tilde{0}1, \tilde{1}0, \tilde{1}1, 0\tilde{0}, 0\tilde{1}, 1\tilde{0}, 1\tilde{1}\}$ is the node set of $G_{3,2}$. Since $f$ (respectively, $f^{-1}$) is the inverse of $f^{-1}$ (respectively $f$), and the inverse of $g_i$ equals $g_{k-i-1}$ for $1 \leq i \leq k-2$, the edges of $G_{k,n}$ are bidirectional. Moreover, according to Definition 1, it is clear that $G_{k,n}$ belongs to Cayley graphs.

**Proposition 1.** *$G_{k,n}$, $n \geq 2$ and $k \geq 3$, is a regular graph of node-degree $k$, and has $\frac{kn(k-1)^n}{2}$ edges.*

*Proof.* Straightforward. □

P. Vadapalli and P. K. Srimani [19] defined the *trivalent Cayley graphs* $TC_n$ as follows.

**Definition 2.** [19] Each node of a *trivalent Cayley graph $TC_n$*, $n \geq 2$, corresponds to a circular permutation in lexicographic order of $n$ symbols, $t_1, t_2, \ldots, t_n$, complemented or uncomplemented. The node set of $TC_n$ is defined as $V(TC_n) = \{t_j^* t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-1}^* | t_i^* = t_i$ or $\bar{t}_i$ for all $i, j \in \{1, \ldots, n\}\}$. Let $\bar{\bar{j}} = j$. Each edge is of the type $(v, \delta(v))$, where $\delta \in \{f_{TC}, f_{TC}^{-1}, g_{TC}\}$, defined in the following way:

$$f_{TC}(t_j^* t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-1}^*) = t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-1}^* \bar{t}_j^*$$
$$f_{TC}^{-1}(t_j^* t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-1}^*) = \bar{t}_{j-1}^* t_j^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-2}^*$$
$$g_{TC}(t_j^* t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots t_{j-1}^*) = t_j^* t_{j+1}^* \ldots t_n^* t_1^* t_2^* \ldots \bar{t}_{j-1}^*$$

We can use English alphabets for representing symbols of $TC_n$. For example, when $n = 4$, $t_1 = a, t_2 = b, t_3 = c$, and $t_4 = d$ with lexicographic order $a < b < c < d$.

Two graphs $G_1$ and $G_2$ are *isomorphic* if there is a one-to-one function $\pi$ from $V(G_1)$ onto $V(G_2)$ such that $(u, v) \in E(G_1)$ if and only if $(\phi(u), \phi(v)) \in E(G_2)$ [23]. We next show that $k$-valent graphs contain trivalent Cayley graphs as its subclass.

**Theorem 2.** *$G_{3,n}$ and $TC_n$ are isomorphic.*

*Proof.* Let $t_1, t_2, \ldots, t_n$ be a lexicographic order of $n$ symbols for $TC_n$. For each node $s_0 s_1 \ldots s_{m-1} \tilde{s_m} s_{m+1} \ldots s_{n-1}$ of $G_{3,n}$, where $s_i \in \{0, 1\}$, we define a function $\pi$ mapping $V(G_{3,n})$ to $V(TC_n)$ as follows:

$$\pi(s_0 s_1 \ldots s_{m-1} \tilde{s_m} s_{m+1} \ldots s_{n-1}) = t'_{n-m+1} t'_{n-m+2} \ldots t'_n t'_1 \ldots t'_{n-m}, \quad (1)$$

where $t'_i = t_i$ if $s_{(m+i-1) \bmod n} = 0$ and $t'_i = \bar{t}_i$ if $s_{(m+i-1) \bmod n} = 1$, for $i \in \{1, \ldots, n\}$. The function $\pi$ is obviously one-to-one and onto.

Suppose that $u^m = s_0^* s_1^* \ldots s_{n-1}^*$ and $v^{m'}$ are two adjacent nodes in $G_{3,n}$. Let $\pi(u^m) = t'_{n-m+1} t'_{n-m+2} \ldots t'_n t'_1 \ldots t'_{n-m}$ according to Equation 1. There are three cases.

**Case 1.** $f(u^m) = v^{m'}$. Then, $v^{m'} = v^{(m-1) \bmod n} = s_1^* s_2^* \ldots s_{n-1}^* \alpha^*$, where $\alpha = (s_0 + 1) \bmod 2$. Then, $\pi(v^{(m-1) \bmod n}) = t'_{n-m+2} \ldots t'_n t'_1 \ldots t'_{n-m} \overline{t'_{n-m+1}}$. Thus, $\pi(u^m)$ and $\pi(v^{m'})$ are adjacent in $TC_n$ because $f_{TC}(\pi(u^m)) = \pi(v^{m'})$.

**Case 2.** $f^{-1}(u^m) = v^{m'}$. Then, $v^{m'} = v^{(m+1) \bmod n} = \beta^* s_0^* s_1^* \ldots s_{n-2}^*$, where $\beta = (\overline{s_{n-1}} - 1) \bmod 2$. So $\pi(v^{(m+1) \bmod n}) = t'_{n-m}t'_{n-m+1} \ldots t'_n t'_1 \ldots t'_{n-m-1}$. Thus, $\pi(u^m)$ and $\pi(v^{m'})$ are adjacent in $TC_n$ because $f_{TC}^{-1}(\pi(u^m)) = \pi(v^{m'})$.

**Case 3.** $g_1(u^m) = v^{m'}$. Then, $v^{m'} = v^m = s_0^* s_1^* s_2^* \ldots \gamma^*$, where $\gamma = (s_{n-1} + 1) \bmod 2$. So $\pi(v^{m'}) = t'_{n-m+1}t'_{n-m+2} \ldots t'_n t'_1 \ldots t'_{n-m-1}\overline{t'_{n-m}}$. Thus, $\pi(u^m)$ and $\pi(v^{m'})$ are adjacent in $TC_n$ because $g_{TC}(\pi(u^m)) = \pi(v^{m'})$.

On the other hand, if $\pi(u^m)$ and $\pi(v^{m'})$ are adjacent in $TC_n$, then $f_{TC}(\pi(u^m)) = \pi(v^{m'})$ or $f_{TC}^{-1}(\pi(u^m)) = \pi(v^{m'})$ or $g_{TC}(\pi(u^m)) = \pi(v^{m'})$. With the proofs similar to that of cases 1–3, we can show the following results to ensure that $u^m$ and $v^{m'}$ are adjacent in $G_{3,n}$: (1) If $f_{TC}(\pi(u^m)) = \pi(v^{m'})$, then $f(u^m) = v^{m'}$; (2) If $f_{TC}^{-1}(\pi(u^m)) = \pi(v^{m'})$, then $f^{-1}(u^m) = v^{m'}$; (3) If $g_{TC}(\pi(u^m)) = \pi(v^{m'})$, then $g_1(u^m) = v^{m'}$. Therefore, since $(u^m, v^{m'}) \in E(G_{3,n})$ if and only if $(\pi(u^m), \pi(v^{m'})) \in E(TC_n)$, $G_{3,n}$ and $TC_n$ are isomorphic. We complete the proof. □

## 3 Shortest path routing and diameter

A graph is said to be *simple* if it contains no loops and multiple edges. A *path* is a simple graph whose vertices can be ordered so that two vertices are adjacent if and only if they are consecutive in the list. Define the distance of $u$ and $v$ to be the length of a shortest path between $u$ and $v$. Since $G_{k,n}$ is a Cayley graph, it is node-symmetric [2]. Thus the distance between two arbitrary nodes is equal to that between some node (source node) and the identity node $\tilde{0}0\ldots0$. Our shortest path routing algorithm aims at constructing a shortest path from a given source node to the identity node. We begin with the following definitions.

For $v = s_0^* s_1^* \ldots s_{n-1}^*$, let $v[i] = s_i^*$, $0 \le i \le n-1$, and let $v[i,j] = s_i^* s_{i+1}^* \ldots s_j^*$, $0 \le i < j \le n-1$. If we view $v = s_0^* s_1^* \ldots s_{n-1}^*$ as a string, then $v[i,j]$ is a *substring.* A 0-*substring* is a substring whose symbols are all zero.

**Definition 3.** Consider an arbitrary node $v = s_0 s_1 \ldots s_{m-1} \tilde{s_m} s_{m+1} \ldots s_{n-1}$ in $G_{k,n}$. The marked symbol $\tilde{s_m}$ divides the string into two parts: the *left part* $s_0 s_1 \ldots s_{m-1}$ and the *right part* $\tilde{s_m} s_{m+1} \ldots s_{n-1}$. We also define the following notations for $v$:

- $NSL_v[x] = |\{s_i : s_i \equiv x \pmod{k-1} \text{ for } 0 \le i \le m-1\}|$.
- $NSR_v[x] = |\{s_i : s_i \equiv x \pmod{k-1} \text{ for } m \le i \le n-1\}|$.
- Let $LZL_v$ be the length of a longest 0-substring in the left part, i.e. $LZL_v = \max\{l \ge 0 : s_j = s_{j+1} = \cdots = s_{j+l-1} = 0,\ 0 \le j \le m-1\}$. Also let $j_{L_v}$ indicate the position of the left-end of the leftmost longest 0-substring in the left part. In particular, define $j_{L_v} = 0$ if $LZL_v = 0$.
- Let $LZR_v$ be the length of a longest 0-substring in the right part, i.e. $LZR_v = \max\{l \ge 0 : s_j = s_{j+1} = \cdots = s_{j+l-1} = 0,\ m \le j \le n-1\}$. Also let $j_{R_v}$ indicate the position of the left-end of the leftmost longest 0-substring in the right part. In particular, define $j_{R_v} = m$ if $LZR_v = 0$.

Throughout this paper, the subscript $v$ can be omitted from the notations $NSL_v[x], NSR_v[x], LZL_v, LZR_v, j_{L_v}$, and $j_{R_v}$ if no ambiguity aries.

For example, consider a node $100\tilde{2}100$ of $G_{4,7}$. Then, $NSL[0] = NSR[0] = 2$; $NSL[1] = NSR[1] = 1$; $NSL[2] = 1; LZL = LZR = 2$; $j_L = 1$ and $j_R = 5$.

We next define the distance function that is useful for our routing algorithm.

**Definition 4.** Consider an arbitrary node $v$ in $G_{k,n}$. Define:

1. $DIST_1(v) = \begin{cases} 2n + m - 2LZL - NSL[0] - NSR[1] - 2 & \text{if } (m - j_L - LZL) > 0 \\ 2n + m - 2LZL - NSL[0] - NSR[1] & \text{otherwise.} \end{cases}$

2. $DIST_2(v) = \begin{cases} 3n - m - 2LZR - NSR[0] - NSL[-1] - 2 & \text{if } (n - j_R - LZR) > 0 \\ 3n - m - 2LZR - NSR[0] - NSL[-1] & \text{otherwise.} \end{cases}$

3. $DIST_3(v) = 2n + m - NSL[-2] - NSR[-1]$.

4. $DIST_4(v) = 3n - m - NSR[2] - NSL[1]$.

The *distance function of* $v$ is defined as $DIST(v) = \min_{1 \le i \le 4}\{DIST_i(v)\}$.

The four terms $DIST_i(v)$'s, $1 \le i \le 4$, defined in Definition 4 represent lengths of four possible paths of $G_{k,n}$ from $v$ to the identity node. Those paths of lengthes $DIST_i(v)$, $1 \leq i \leq 4$, are constructed using algorithms DIST_1–DIST_4, respectively. As it will be shown later, the distance function $DIST(v)$ computes the length of a shortest path from $v$ to the identity node. For ease of understanding our shortest path routing algorithm, we first present Algorithm DIST_1 to generate a path from $v$ to the identity node with the length $DIST_1(v)$.

**Algorithm 1** DIST_1$(v, n, m, LZL, j_L)$

**for** $i = 0$ to $(j_L - 1)$ **do**
  $v \leftarrow f(v)$
**end for**
**for** $i = 0$ to $(j_L - 1) + (n - m)$ **do**
  **if** $v[n-1] \neq 1$ **then**
    $v \leftarrow g_{k-v[n-1]}(v)$
  **end if**
  $v \leftarrow f^{-1}(v)$
**end for**
**if** $(m - j_L - LZL) > 0$ **then**
  **for** $i = 1$ to $(m - j_L - LZL - 1)$ **do**
    $v \leftarrow f^{-1}(v)$
  **end for**
  $v \leftarrow g_{k-v[n-1]-1}(v)$
  **for** $i = 1$ to $(m - j_L - LZL - 1)$ **do**
    $v \leftarrow f(v)$
    **if** $v[n-1] \neq 0$ **then**
      $v \leftarrow g_{k-v[n-1]-1}(v)$
    **end if**
  **end for**
**end if**

Since the concepts of algorithms DIST_2–DIST_4 are similar to that of Algorithm DIST_1, we presented them in Appendix. Below is our shortest path routing algorithm.

**Algorithm 2** OPTMAL_ROUTE

compute the value $l \in \{1, 2, 3, 4\}$ such that $DIST_l(v)$ is the minimum of $DIST_i(v)$'s for all $1 \leq i \leq 4$.
execute Algorithm DIST_$l$

The following lemma can help in proving the correctness of our routing algorithm.

**Lemma 3.** *For an arbitrary node $v$ in $G_{k,n}$, $DIST(v) - DIST(v') = \pm 1$ or $0$, where $v' = \delta(v)$ and $\delta \in \{f, f^{-1}, g_1, g_2, \ldots, g_{k-2}\}$.*

*Proof.* Assume that $v = s_0 s_1 \ldots s_{m-1} \tilde{s_m} s_{m+1} \ldots s_{n-1}$ and $\delta(v) = v' = t_0 t_1 \ldots t_{m'-1} \tilde{t_{m'}} t_{m'+1} \ldots t_{n-1}$. Let $NSL'$, $NSR'$, $LZL'$, and $LZR'$ be the corresponding parameters (defined in Definition 3) of $v'$. By the definition of the distance function, we can utilize the fact that $\min_{1 \leq i \leq 4}\{DIST_i(v)\} - DIST_p(v') \leq DIST(v) - DIST(v') \leq DIST_q(v) - \min_{1 \leq i \leq 4}\{DIST_i(v')\}$, for some $1 \leq p, q \leq 4$, to simplify the computation.

**Case 1.** $\delta \in \{g_1, g_2, \ldots, g_{k-2}\}$. In this case, we have that $m' = m$, $NSL'[x] = NSL[x]$, and $LZL' = LZL$. According to Definition 4, the following results can be obtained.

**(1)** $DIST_1(v) - DIST_1(v') = -NSR[1] + NSR'[1] = \pm 1$ or $0$.

**(2)** Let $q = -2LZR - NSR[0] + 2LZR' + NSR'[0]$. According to Definition 4, we have the following results:

**i.** If $v[n-1] \neq 0$ and $v'[n-1] \neq 0$, then $q = 0$ and $DIST_2(v) - DIST_2(v') = q = 0$;
**ii.** If $v[n-1] \neq 0, v'[n-1] = 0$, and $LZR = LZR'$, then $q = 1$ and $DIST_2(v) - DIST_2(v')$ may equal $q = 1$;
**iii.** If $v[n-1] \neq 0, v'[n-1] = 0$, and $LZR \neq LZR'$, then $q = 3$ and $DIST_2(v) - DIST_2(v') = q - 2 = 1$;
**iv.** If $v[n-1] = 0, v'[n-1] \neq 0$, and $LZR = LZR'$, then $q = -1$ and $DIST_2(v) - DIST_2(v') = q = -1$;
**v.** If $v[n-1] = 0, v'[n-1] \neq 0$, and $LZR \neq LZR'$, then $q = -3$ and $DIST_2(v) - DIST_2(v') = q + 2 = -1$.

**(3)** $DIST_3(v) - DIST_3(v') = -NSR[-1] + NSR'[-1] = \pm 1$ or $0$.

**(4)** $DIST_4(v) - DIST_4(v') = -NSR[2] + NSR'[2] = \pm 1$ or $0$.

Combining (1)–(4), we have that $DIST(v) - DIST(v') = \pm 1$ or $0$.

**Case 2.** $\delta = f$.

**Case 2.1.** $m \neq 0$. In this case, we have that $m' = m - 1$.

**(1)** By definition 4, $DIST_1(v) - DIST_1(v') = 1 - 2LZL - NSL[0] - NSR[1] + 2LZL' + NSL'[0] + NSR'[1]$. Then, $DIST_1(v) - DIST_1(v') =$
$$\begin{cases} 1 - 2LZL + 2LZL' = \pm 1 & \text{if } v[0] = 0; \\ 1 & \text{otherwise.} \end{cases}$$

**(2)** Let $q = -1 - 2LZR - NSR[0] - NSL[-1] + 2LZR' + NSR'[0] + NSL'[-1]$. By definition 4, $DIST_2(v) - DIST_2(v')$ may equal $q$ or $q - 2$ or $q + 2$ in this case. Then, $DIST_2(v) - DIST_2(v') =$
$$\begin{cases} -1 - 2LZR + 2LZR' = \pm 1 & \text{if } v[0] = k - 2; \\ -1 & \text{otherwise.} \end{cases}$$

**(3)** By Definition 4, $DIST_3(v) - DIST_3(v') = 1 - NSL[-2] - NSR[-1] + NSL'[-2] + NSR'[-1]$. By discussing the situations that $v[0] = k-3$ or $v[0] \neq k-3$, we conclude that $DIST_3(v) - DIST_3(v') = 1$.

**(4)** By Definition 4, $DIST_4(v) - DIST_4(v') = -1 - NSR[2] - NSL[1] + NSR'[2] + NSL'[1]$. By discussing the situations that $v[0] = 1$ or $v[0] \neq 1$, we have that $DIST_4(v) - DIST_4(v') = -1$.

Combining (1)–(4), we have that $DIST(v) - DIST(v') = \pm 1$ or 0.

**Case 2.2.** $m = 0$. In this case, we have that $m' = n-1$, $NSL[x] = 0$, and $LZL = 0$. Thus, $DIST_1(v) = 2n - NSR[1]$.

$$DIST_2(v) = \begin{cases} 3n - 2LZR - NSR[0] - 2 \\ \quad \text{if } (n - j_R - LZR) > 0; \\ 3n - 2LZR - NSR[0] \\ \quad \text{otherwise.} \end{cases}$$

$DIST_3(v) = 2n - NSR[-1]$.
$DIST_4(v) = 3n - NSR[2] \geq 2n$.
Note that we do not need to consider $DIST_4(v)$ in computing $DIST(v)$. On the other hand, we have that

$$DIST_1(v') = \begin{cases} 3n - 1 - 2LZL' - NSL'[0] - NSR'[1] - 2 \\ \quad \text{if } (m' - j_{L_{v'}} - LZL') > 0; \\ 3n - 1 - 2LZL' - NSL'[0] - NSL'[1] \\ \quad \text{otherwise.} \end{cases}$$

$$DIST_2(v') = \begin{cases} 2n + 1 - 2LZR' - NSR'[0] - NSL'[-1] - 2 \\ \quad \text{if } (n - j_{R_{v'}} - LZR') > 0; \\ 2n + 1 - 2LZR' - NSR'[0] - NSL'[-1] \\ \quad \text{otherwise.} \end{cases}$$

$DIST_3(v') = 3n - 1 - NSL'[-2] - NSR'[-1] \geq 2n - 1 \geq DIST_2(v')$ (this term can be ignored in computing $DIST(v')$);
and $DIST_4(v') = 2n + 1 - NSR'[2] - NSL'[1]$.
Then, the following results hold:

**(1)** Since $DIST_1(v) - DIST_4(v') = -1 - NSR[1] + NSR'[2] + NSL'[1]$, we can show that $DIST_1(v) - DIST_4(v') = -1$ without regarding whether $v[0] = 1$ or not.

**(2)** Since $DIST_2(v) - DIST_1(v') = 1 - 2LZR - NSR[0] + NSR'[1] + 2LZL' + NSL'[0]$, we have that $DIST_2(v) - DIST_1(v') =$

$$\begin{cases} \pm 1 & \text{if } v[0] = 0; \\ 1 & \text{otherwise.} \end{cases}$$

**(3)** Clearly, $DIST_3(v) - DIST_2(v') =$

$$\begin{cases} -1 - NSR[-1] + 2LZR' + NSR'[0] \\ \quad + NSL'[-1] \quad \text{if } v[0] = k-2; \\ 1 - NSR[-1] + 2LZR' + NSR'[0] \\ \quad + NSL'[-1] \quad \text{otherwise.} \end{cases}$$

It is not difficult to verify that $DIST_3(v) - DIST_2(v') = 1$ without regarding whether $v[0] = k-2$ or not.

From (1)–(3), $DIST(v) - DIST(v') = \pm 1$ or 0.

**Case 3.** $\delta = f^{-1}$. Then, $v' = f^{-1}(v) \Leftrightarrow f(v') = v$. This is symmetric to Case 2.

□

**Lemma 4.** *For the identity node $I$ in $G_{k,n}$, $DIST(I) = 0$ and $DIST(\delta(I)) = 1$ for any $\delta \in \{f, f^{-1}, g_1, g_2, \ldots, g_{k-2}\}$.*

*Proof.* Straightforward. □

**Theorem 5.** *Given an arbitrary node $v$ in $G_{k,n}$, Algorithm* OPTIMAL_ROUTE *correctly generates an optimal (shortest) path of length $DIST(v)$ from $v$ to the identity node.*

*Proof.* Clearly, Algorithm OPTIMAL_ROUTE constructs a path of length $DIST(v)$ for a given node $v$. By Lemma 3, since an application of any one generator can decrease the value of $DIST(v)$ at best by one, the algorithm does construct a shortest path from $v$ to the identity node. □

**Corollary 6.** *For an arbitrary node $v$ in $G_{k,n}$, $DIST(v)$ equals the distance between $v$ and the identity node.*

*Proof.* It follows directly from Theorem 5. □

We can utilize our shortest path routing algorithm to compute the diameter $diam(G_{k,n})$ for $k \geq 6$.

**Theorem 7.** *For an arbitrary node $v$ in $G_{k,n}$, $DIST(v) \leq \max\{2n, \frac{5n}{2} - 2\}$. Moreover, $diam(G_{k,n}) = \lfloor \frac{5n}{2} \rfloor - 2$ when $k \geq 6$ and $n \geq 4$; and $diam(G_{k,n}) = 2n$ when $k \geq 6$ and $n = 2, 3$.*

*Proof.* By Definition 4, we have that $DIST(v) \leq \min\{2n + m - 2, 3n - m - 2, 2n + m, 3n - m\} = \min\{2n + m - 2, 3n - m - 2\} \leq \frac{5n}{2} - 2$ for $m > 0$, and $DIST(v) \leq \min\{2n, 3n - 2\} \leq 2n$ for $m = 0$. Hence the upper bound of the diameter is $\max\{\frac{5n}{2} - 2, 2n\}$. Note that it is $\frac{5n}{2} - 2$ for $n \geq 4$.

We next show the lower bound. For $k \geq 6$ and $n \geq 4$, consider the node $v$ in $G_{k,n}$ such that $v[i] \notin \{0, 1, k-2, k-3\}$ and $m = \lfloor n/2 \rfloor$. Thus $NSL[0] = NSL[1] = NSL[-1] = NSL[-2] = NSR[0] = NSR[1] = NSR[-1] = NSR[-2] = 0$ and $LZL = LZR = 0$. By Definition 4, $DIST(v) = \min\{2n + \lfloor \frac{n}{2} \rfloor - 2, 3n - \lfloor \frac{n}{2} \rfloor - 2\} = \lfloor \frac{5n}{2} \rfloor - 2$. Therefore, $\lfloor \frac{5n}{2} \rfloor - 2 \leq diam(G_{k,n}) \leq \frac{5n}{2} - 2$ and $diam(G_{k,n}) = \lfloor \frac{5n}{2} \rfloor - 2$. The case of $k \geq 6$ and $n = 2, 3$ can be shown similarly. □

## 4 Embedding

In this section, we show that the $k$-valent graph can embed two types of structures, cycles and cliques.

A *cycle* is a graph with an equal number of nodes and edges whose nodes can be placed around a circle so that two vertices are adjacent if and only if they appear consecutively along the circle [23]. Two cycles are *node-disjoint* if they have no common node. An edge defined by the generator $f$ (or $f^{-1}$) is an $\mathcal{F}$*-edge*. Any cycle in $G_{k,n}$ consisting of only $\mathcal{F}$-edges is an $\mathcal{F}$*-cycle*. For example, the cycle $(\tilde{0}0, 0\tilde{1}, \tilde{1}1, 1\tilde{0})$ in $G_{3,2}$ is an $\mathcal{F}$-*cycle*. For non-negative integer $i$, we recursively define $f^i(v) = \begin{cases} v & \text{if } i = 0, \\ f(f^{i-1}(v)) & \text{if } i > 0. \end{cases}$ The notation $f^{-i}(v)$ can be defined similarly.

**Theorem 8.** *$G_{k,n}$ can embed $(k-1)^{n-1}$ node-disjoint $\mathcal{F}$-cycles of length $n(k-1)$.*

*Proof.* Given an arbitrary node $v$ in $G_{k,n}$, it is not difficult to verify that $f^{n(k-1)}(v) = v$ and $f^i(v) \neq f^j(v)$, where $1 \leq i, j \leq n(k-1)$ and $i \neq j$. Thus, from an arbitrary node $v$, a cycle of length $n(k-1)$ in $G_{k,n}$ can be generated by a functional iteration $f^{n(k-1)}(v)$. Since $G_{k,n}$ has $n(k-1)^n$ nodes, there are totally $(k-1)^{n-1}$ $\mathcal{F}$-cycles of each length $n(k-1)$ can be generated. These cycles are node-disjoint by the fact that $f(u) = f(v)$ if and only if $u = v$. □

Each $\mathcal{F}$-cycle has a unique node $v$ such that $v[i] = s$ and $m = 0$ for two given integers $0 \leq i \leq n-1$ and $0 \leq s \leq k-2$. Therefore, for each $\mathcal{F}$-cycle $C$, we can define $C$*-leader* (*leader* for short if no ambiguity aries) to be the unique node $v$ of $C$ satisfying that $v[0] = 0$ and $m = 0$. Note that each leader uniquely determines a specific $\mathcal{F}$-cycle. Two $\mathcal{F}$-cycles $C_1$ and $C_2$ are said to be *adjacent* if there exists a node $v_1 \in C_1$ and a node $v_2 \in C_2$ such that $v_1 = g_i(v_2)$ for some $1 \leq i \leq k-2$.

**Theorem 9.** *Each $\mathcal{F}$-cycle of $G_{k,n}$ is adjacent to $n(k-2)$ (respectively, $k-2$) different $\mathcal{F}$-cycles for $n > 2$ (respectively, $n = 2$).*

*Proof.* We first consider $n > 2$. For an arbitrary $\mathcal{F}$-cycle $C$ with the $C$-leader $v = \tilde{0}s_1s_2 \ldots s_{n-1}$,

$$f^{-i} \circ (g_j \circ f^i(v)) \quad = \quad \tilde{0}s_1s_2 \ldots s' \ldots s_{n-1}, \quad (2)$$

where $\circ$ means function composition and $s' = (s_{i-1}+j) \bmod (k-1)$ for all $i \neq 1$ and $1 \leq j \leq k-2$. Thus $(n-1)(k-2)$ leaders of $(n-1)(k-2)$ distinct $\mathcal{F}$-cycles can be arrived from $v$. This implies that the given $\mathcal{F}$-cycle is adjacent to $(n-1)(k-2)$ distinct $\mathcal{F}$-cycles. Moreover, the other $(k-2)$ leaders can be arrived from $v$ by

$$f^{-1-jn} \circ (g_j \circ f(v)) \quad = \quad \tilde{0}s_1's_2' \ldots s_{n-1}', \quad (3)$$

where $s_i' = (s_i - j) \bmod (k-1)$. Therefore, each $\mathcal{F}$-cycle is adjacent to $n(k-2)$ different $\mathcal{F}$-cycles.

The case of $n = 2$ can be shown using either Equation 2 or Equation 3. □

The proof of the above theorem leads to the following result.

**Corollary 10.** *An $\mathcal{F}$-cycle with its leader $v = \tilde{0}x_1x_2 \ldots x_{n-2}$ is adjacent to following $\mathcal{F}$-cycles with leaders $\tilde{0}y_1y_2 \ldots y_{n-1}$ such that one of the following conditions hold.*

1. *$x_i \neq y_i$ for some $1 \leq i \leq n-1$, and $x_j = y_j$ for all $1 \leq j \leq n-1$ and $j \neq i$;*
2. *$x_1 - y_1 \equiv x_2 - y_2 \equiv \cdots \equiv x_{n-1} - y_{n-1} \pmod{k-1}$.*

We next show that $G_{k,n}$ can embed cliques. An edge of $G_{k,n}$ defined by the generator $g_i$ is called $\mathcal{G}$*-edge*. A *clique* in a graph is a set of pairwise adjacent nodes. A clique is said to be *maximal* if it is not contained in another clique. A $\mathcal{G}$*-clique* is a maximal clique in $G_{k,n}$ such that each pair of adjacent nodes is connected by a $\mathcal{G}$-edge. For example, $\{\tilde{0}0, \tilde{0}1\}$ is a $\mathcal{G}$-clique in $G_{3,2}$. Note that a $\mathcal{G}$-clique contains $k-1$ nodes. Given some fixed $0 \leq s \leq k-2$, each $\mathcal{G}$-clique has a unique node $v$ satisfying that $v[n-1] = s$. Therefore, for a $G$-clique $K$, we can define the $K$*-leader* (*leader* for short if no ambiguity aries) to be the unique node $v \in K$ satisfying that $v[n-1] = 0$. We use such a node to represent its corresponding $\mathcal{G}$-clique for convenience. Two $\mathcal{G}$-cliques are *node-disjoint* if they have no common node.

**Theorem 11.** *$G_{k,n}$ can embed $n(k-1)^{n-1}$ node-disjoint $\mathcal{G}$-cliques of size $k-1$.*

*Proof.* Clearly, there are $n(k-1)^{n-1}$ leaders in $G_{k,n}$, which correspond to $n(k-1)^{n-1}$ $\mathcal{G}$-cliques. It is not difficult to verify that these $\mathcal{G}$-cliques are pairwise node-disjoint. Moreover, since each $\mathcal{G}$-clique contains $k-1$ nodes, the $n(k-1)^{n-1}$ $\mathcal{G}$-cliques form a partition of the vertices of $G_{k,n}$. □

Two $\mathcal{G}$-cliques $K_1$ and $K_2$ are said to be *adjacent* if there exists a pair of vertices $u \in K_1$ and $v \in K_2$ such that either $v = f(u)$ or $v = f^{-1}(u)$.

**Theorem 12.** *Each $\mathcal{G}$-clique in $G_{k,n}$ is adjacent to $2(k-1)$ (respectively, $k-1$) different $\mathcal{G}$-cliques when $n > 2$ (respectively, $n = 2$).*

*Proof.* We first consider that $n > 2$. Given an arbitrary $\mathcal{G}$-clique with its leader $v = s_0^* s_1^* \ldots s_{n-2}^* 0^*$, we can apply $f^i \circ g_j(v)$ for $i = \pm 1$ and $1 \leq j \leq k-2$, to reach $2(k-2)$ distinct $\mathcal{G}$-cliques, and apply $f^{\pm 1}(v)$ to reach another two distinct $\mathcal{G}$-cliques. Therefore, each $\mathcal{G}$-clique is adjacent to $2(k-1)$ distinct $\mathcal{G}$-cliques.

When $n = 2$, we can apply $f^1 \circ g_j(v)$ and $f^1(v)$ (or $f^{-1} \circ g_j(v)$ and $f^{-1}(v)$) to reach $k-1$ distinct $\mathcal{G}$-cliques. Note that the $\mathcal{G}$-cliques reached by $\{f^1 \circ g_j(v), f^1(v)\}$ are the same with those reached by $\{f^{-1} \circ g_j(v), f^{-1}(v)\}$. □

## 5 Maximally fault tolerance

The node fault tolerance of an undirected graph is measured by the node-connectivity of the graph [24]. The node-connectivity of a graph $G$, written $\kappa(G)$, is the minimum size of a node set $S$ such that $G - S$ is disconnected or has only one node [23]. A graph $G$ is $\xi$-connected if its node-connectivity is at least $\xi$. Obviously, the node-connectivity of a graph $G$ cannot exceed the minimum degree of a node in $G$; thus $\kappa(G_{k,n}) \leq k$ since $G_{k,n}$ is $k$-regular. A graph is said to be *maximally fault-tolerant* if node-connectivity equals the minimum node-degree of the given graph. A set of paths from $u$ to $v$ are *pairwise internally disjoint* if any two of them have no common internal nodes. In this section, we show that $\kappa(G_{k,n}) = k$ and hence this class of graphs is maximally fault-tolerant.

**Lemma 13.** [19] *$\kappa(RG_{3,n}) = 3$ for $n \geq 2$.*

Since the node-connectivity of $G_{k,n}$ for $k = 3$ was shown in [19], we only consider $k \geq 4$ in the reminder of this section.

**Definition 5.** For a $k$-valent graph $G_{k,n}$, the *reduced graph $RG_{k,n}$* of $G_{k,n}$ can be constructed using the following two steps:

1. For each $\mathcal{F}$-cycle with its leader $\tilde{0}s_1 s_2 \ldots s_{n-1}$, shrink it into a single node, namely *super node*, and label it with $s_1 s_2 \ldots s_{n-1}$;

2. Connect two arbitrary super nodes by an undirected edge, namely *super edge*, if the corresponding $\mathcal{F}$-cycles are adjacent in $G_{k,n}$.

For convenience, we use the notation $P[x, y]$ to denote a path from $x$ to $y$.

**Lemma 14.** *$\kappa(RG_{k,n}) \geq k$ for $k \geq 4$ and $n \geq 3$.*

*Proof.* Consider a subgraph $R_{k,n}$ of $RG_{k,n}$ defined as follows: The node set of $R_{k,n}$ is the same with $RG_{k,n}$; and two node $x_1 x_2 \ldots x_{n-1}$ and $y_1 y_2 \ldots y_{n-1}$ of $R_{k,n}$ are adjacent iff there is exactly one $i$ such that $x_i \neq y_i$ and $x_j = y_j$ for all $j \neq i$.

We show the lemma by showing that there are $k$ pairwise internally disjoint paths between two arbitrary distinct nodes in $R_{k,n}$. The proof is by induction on $n$.

Basis: Prove that the result is true for $n = 3$. Consider two arbitrary nodes $u$ and $v$ in $R_{k,3}$.

**Case 1.** $u$ and $v$ have exactly one different symbol. Without loss of generality, assume that $u = 0x$ and $v = 0y$. According to Corollary 10 and the construction of $R_{k,n}$, $k$ pairwise internally disjoint paths from $u$ to $v$ can be constructed as follows: $(0x, 0y)$, $(0x, 1x, 1y, 0y)$, $\ldots$, $(0x, (k-2)x, (k-2)y, 0y)$ and $(0x, 0z, 0y)$ for $z \in \{0, 1, \ldots, k-2\} \backslash \{x, y\}$.

**Case 2.** $u$ and $v$ have two different symbols. Assume that $u = u_1 u_2$ and $v = v_1 v_2$. Then, according to Corollary 10 and the construction of $R_{k,n}$, $2k-4$ ($\geq k$ when $k \geq 4$) pairwise internally disjoint paths from $u$ to $v$ can be constructed as follows: $(u_1 u_2, u_1 v_2, v_1 v_2)$, $(u_1 u_2, v_1 u_2, v_1 v_2)$, $(u_1 u_2, x u_2, x v_2, v_1 v_2)$, and $(u_1 u_2, u_1 x, v_1 x, v_1 v_2)$ for all $x \in \{0, 1, \ldots, k-2\} \setminus \{u_1, v_1\}$.

Thus the base case is true.

Induction step: Assume that the result is true for $n-1$ and then show it is true for $n \geq 4$. It is not difficult to verify that $R_{k,n}$ has a recursive structure which can be decomposed into $k-1$ copies of $R_{k,n-1}$. Each $R_{k,n-1}$ is a subgraph of $R_{k,n}$ induced by nodes in $x_1 x_2 \ldots x_{n-2} y$ for some fixed symbol $y$ and $x_i \in \{0, 1, \ldots, k-2\}$. Consider two arbitrary node $u$ and $v$. If $u$ and $v$ are in the same $R_{k,n-1}$, then there are $k$ pairwise internally disjoint paths from $u$ to $v$ by induction hypothesis. Otherwise, $u$ and $v$ are in different $R_{k,n-1}$'s. Assume that $u = u_1 u_2 \ldots u_{n-2} 0$ and $v = v_1 v_2 \ldots v_{n-2} 1$. By the induction hypothesis, there are $k$ pairwise internally disjoint paths $P_1, P_2, \ldots, P_k$ from $u$ to the node $v' = v_1 v_2 \ldots v_{n-2} 0$ because $u$ and $v'$ are in the same $R_{k,n-1}$. Let $n_1, n_2, \ldots, n_k$ be the neighbors of $v'$ which belong to paths $P_1, P_2, \ldots, P_k$, respectively. According to Corollary 10 and all $n_i$'s are in the same $R_{k,n-1}$, we have that $n_i = t_1^i t_2^i \ldots t_{n-2}^i 0$ in which $t_j^i \neq v_j$ for some $j$ and $t_l^i = v_l$ for all $1 \leq l \leq n-2$ and $l \neq j$. Let $n_i'$ be the neighbor of $n_i$ which is located in the same $R_{k,n-1}$ with $v$, that is, $n_i' = t_1^i t_2^i \ldots t_{n-2}^i 1$. Note that $n_i'$, for all $1 \leq i \leq k$, are adjacent to $v$. Also let $P_i[u, n_i]$ be the subpath of $P_i$ from $u$ to $n_i$, $1 \leq i \leq k$. Then, $k$ pairwise internally disjoint paths from $u$ to $v$ can be constructed as $P_i[u, n_i] + (n_i, n_i') + (n_i', v)$ for all $i = 1, 2, \ldots, k$, where " + " means the path-concatenation operation[1]. □

The following lemma is useful to prove Lemma 16.

**Lemma 15.** [12] *Given two sets of nodes $V_1$ and $V_2$ such that $|V_1| = |V_2| = \xi$ in a $\xi$-connected graph, there are $\xi$ node disjoint paths connecting the nodes form $V_1$ to $V_2$.*

**Lemma 16.** $\kappa(G_{k,n}) \geq k$ *for $k \geq 4$ and $n \geq 2$.*

*Proof.* We show the lemma by showing that there are $k$ pairwise internally disjoint paths between two arbitrary nodes $u$ and $v$ in $G_{k,n}$. There are two cases.

**Case 1.** $n = 2$. Recall that in this case, $RG_{k,2}(= R_{k,2})$ is a complete graph on $k-1$ nodes. Thus there are totally $k-1$ $\mathcal{F}$-cycles in $G_{k,2}$.

**Case 1.1.** $u$ and $v$ are in the same $\mathcal{F}$-cycle $C$. This cycle is a concatenation of two internally disjoint paths from $u$ to $v$, denoted by $P_1$ and $P_2$, which contribute two paths. Let $C_1, C_2, \ldots C_{k-2}$ be the other $\mathcal{F}$-cycles of $G_{k,2}$. Then, the other $k-2$ paths can be constructed using the following three steps:

**1.** For each $C_i$, identify the *entry node* $g_{j_i}(u) \in C_i$ and the *exit node* $g_{l_i}(v) \in C_i$, where $1 \leq j_i, l_i \leq k-2$. [2]

**2.** For each $C_i$, construct a path $Q_i$ from $g_{j_i}(u)$ to $g_{l_i}(v)$ using the $\mathcal{F}$-edges of corresponding $\mathcal{F}$-cycle.

**3.** Construct the desired $k$ internally disjoint paths: $P_1$, $P_2$, $(u, g_{j_1}(u)) + Q_1 + (g_{l_1}(v), v)$, $(u, g_{j_2}(u)) + Q_2 + (g_{l_2}(v), v)$, $\ldots$, $(u, g_{j_{k-2}}(u)) + Q_{k-2} + (g_{l_{k-2}}(v), v)$.

**Case 1.2.** $u$ and $v$ are in different $\mathcal{F}$-cycles. Let $F_1$ and $F_2$ be the $\mathcal{F}$-cycles to which $u$ and $v$ belong, respectively. We first construct three pairwise internally disjoint paths from $u$ to $v$ using $F_1$ and $F_2$. If $u$ and $v$ are adjacent, then the desired paths can be constructed as follows:

**A.** Find two nodes $u_1$ and $u_2$ (respectively, $v_1$ and $v_2$) different from $u$ (respectively, $v$) in $F_1$ (respectively, $F_2$) such that $g_i(u_1) = v_1$ and $g_i(u_2) = v_2$ for some $1 \leq i \leq k-2$.

**B.** Let $P[u, u_1]$ and $P[u, u_2]$ (respectively, $Q[v_1, v]$ and $Q[v_2, v]$) be two paths in $F_1$ (respectively, $F_2$) whose nodes are all different but $u$ (respectively, $v$). Then, $(u, v)$, $P[u, u_1] + (u_1, v_1) + Q[v_1, v]$, and $P[u, u_2] + (u_2, v_2) + Q[v_2, v]$ form the desired three paths.

Otherwise, if $u$ and $v$ are not adjacent, then the desired three paths can be constructed as follows:

**I.** Find two nodes $u_1$ and $u_2$ (respectively, $v_1$ and $v_2$) different from $u$ (respectively, $v$) in $F_1$ (respectively, $F_2$) such that $g_i(u_1) = v$, $g_j(u_2) = v_1$, and $g_l(u) = v_2$, where $1 \leq i, j, l \leq k-2$.

**II.** Let $P'[u, u_1]$ and $P'[u, u_2]$ (respectively, $Q'[v_1, v]$ and $Q'[v_2, v]$) be two paths in $F_1$ (respectively, $F_2$) whose nodes are all different but $u$ (respectively, $v$). Then, $P'[u, u_1] + (u_1, v)$, $(u, v_2) + Q'[v_2, v]$, and $P'[u, u_2] + (u_2, v_1) + Q'[v_1, v]$ form the desired three paths.

The other $k-3$ internally disjoint paths from $u$ to $v$ can be constructed from the other $k-3$ $f$-cycles using the method similar to that of Case 1.1.

**Case 2.** $n \geq 3$. If $u$ and $v$ are in the same $\mathcal{F}$-cycle, there exist $k$ internally disjoint paths from $u$ to $v$: two paths are constructed in the same $\mathcal{F}$-cycle, and the other $k-2$ paths are constructed from adjacent $k-2$ $\mathcal{F}$-cycles based on the method similar to that of Case 1. If $u$ and $v$ are in different $\mathcal{F}$-cycles, then the set of nodes $\{g_1(u), g_2(u), \ldots, g_{k-2}(u), g_1(f(u)), g_1(f^{-1}(u))\}$ (respectively, $\{g_1(v), g_2(v), \ldots, g_{k-2}(v), g_1(f(v)), g_1(f^{-1}(v))\}$) which are in $k$ different $\mathcal{F}$-cycles, denoted by $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_k\}$ (respectively, $\{\mathcal{C}'_1, \mathcal{C}'_2, \ldots, \mathcal{C}'_k\}$), and can be reached from $u$ (respectively, $v$). By Lemma 14 and Lemma 15, there are $k$ node-disjoint paths connecting $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_k\}$ and $\{\mathcal{C}'_1, \mathcal{C}'_2, \ldots, \mathcal{C}'_k\}$ in $RG_{k,n}$. This implies that there exist $k$ pairwise internally disjoint paths from $u$ to $v$ in $G_{k,n}$.

□

We conclude this section with the following result.

**Theorem 17.** $\kappa(G_{k,n}) = k$ *for $k \geq 3$ and $n \geq 2$.*

*Proof.* By Lemma 13 and Lemma 16, we have that $\kappa(G_{k,n}) \geq k$. By the fact that $G_{k,n}$ is a $k$-regular graph, we have that $\kappa(G_{k,n}) \leq k$. Thus the result holds. □

[1] In the remainder of this paper, the notation "+" is used to mean the path-concatenation.

[2] It is not difficult to show that there exist $j_i$ and $l_i$ such that $g_{j_i}(u)$ and $g_{l_i}(v)$ are in the same $\mathcal{F}$-cycle $C_i$.

## 6 Concluding remarks

In this paper, we generalize the concept of the trivalent graph and propose a new family of $k$-valent graphs $G_{k,n}$ for designing interconnection networks. The proposed graph is regular with the node-degree $k$, where $k$ can be any integer larger than 3. We also show that $G_{k,n}$ for some fixed integer $k \geq 6$, has logarithmic diameter subject to the number of nodes, and has maximally fault tolerance. Besides, a shortest path routing algorithm is also presented.

We also show that $G_{k,n}$ can embed cycles. A cycle structure, which is a fundamental topology for parallel and distributed processing, is suitable for local area networks and for the development of simple parallel algorithms with low communication cost [17]. A cycle structure can also be used as a control/data flow structure for distributed computation in arbitrary networks. As with our result, $(k-1)^{n-1}$ node-disjoint cycles can be emulated in $G_{k,n}$. Previously, a number of parallel algorithms that can be executed on cycles have been developed [3]. Those algorithms can be executed as well on $G_{k,n}$.

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# A Appendix

**Algorithm 3** DIST_2$(v, n, m, LZR, j_R)$

```
for i = 0 to (n − 2 − j_R − LZR) do
    v ← f^{-1}(v)
end for
if (n − j_R − LZR) > 0 then
    v ← g_{k−v[n−1]−1}(v)
    for i = 0 to (n − 2 − j_R − LZR) do
        v ← f(v)
        if v[n − 1] ≠ 0 then
            v ← g_{k−v[n−1]−1}(v)
        end if
    end for
end if
for i = 0 to (m − 1) do
    v ← f(v)
    if v[n − 1] ≠ 0 then
        v ← g_{k−v[n−1]−1}(v)
    end if
end for
for i = 0 to j_R − m − 1 do
    v ← f(v)
end for
for i = 0 to j_R − m − 1 do
    if v[n − 1] ≠ 1 then
        v ← g_{k−v[n−1]}(v)
    end if
    v ← f^{-1}(v)
end for
```

**Algorithm 4** DIST_3$(v, n, m)$

```
for i = 0 to (m − 1) do
    v ← f(v)
    if v[n − 1] ≠ k − 2 then
        v ← g_{k−2−v[n−1]}(v)
    end if
end for
for i = m to (n − 1) do
    v ← f(v)
    if v[n − 1] ≠ 0 then
        v ← g_{k−1−v[n−1]}(v)
    end if
end for
for i = 0 to (m − 1) do
    v ← f(v)
end for
```

**Algorithm 5** DIST_4$(v, n, m)$

```
for i = m to (n − 1) do
    if v[n − 1] ≢ 2 (mod k − 1) then
        v ← g_{k−v[n−1]+1}(v)
    end if
    v ← f^{-1}(v)
end for
for i = 0 to (m − 1) do
    if v[n − 1] ≠ 1 then
        v ← g_{k−v[n−1]}(v)
    end if
    v ← f^{-1}(v)
end for
for i = m to (n − 1) do
    v ← f^{-1}(v)
end for
```